THE NOVA RESEARCH CURRICULUM

Mathematics

Grade 3 · Term 1

Student edition

Term 1 outcomes: MA-3.1 numbers to 10,000 and rounding, MA-3.2 four-digit addition and subtraction, MA-3.3 (part) tables of 6 and 7 and multiplying a two-digit by a one-digit number. Six lessons a week.

Lesson notes

MA-3.1 Numbers to 10,000; place value; rounding; odd and even; Roman numerals (Weeks 1–4)

1. **Place value (Week 1):** Th | H | T | O chart with base-10 blocks or place-value counters; 4,732 = 4 thousands, 7 hundreds, 3 tens, 2 ones.
2. **Compare and order (Week 2):** use <, > and =; compare from the biggest place first.
3. **Rounding (Week 3):** use a number line: 4,732 is nearer 4,730 than 4,740; nearer 4,700 than 4,800. Rule: 5 or more rounds up.
4. **Odd and even, Roman numerals to 20 (Week 4):** even numbers end in 0, 2, 4, 6, 8; I, V, X and how IV and IX work.

Watch out: mixing up 4,006 and 4,060, and leaving out zeros that hold a place. Always use the chart.

MA-3.2 Adds and subtracts up to four digits; mental sums within 100 (Weeks 5–8)

1. **Estimate first:** round both numbers to the nearest 100 and estimate the answer.
2. **Column method:** line up places; carry and borrow across up to three places.
3. **Borrowing across zeros:** 6,000 − 2,345 with counters before the written method.
4. **Check:** add the answer back to test subtraction; compare with the estimate.
5. **Mental (5 min daily):** 47 + 38 as 47 + 40 − 2.

MA-3.3 (part) Tables of 6 and 7; two-digit × one-digit (Weeks 9–12)

1. **Build the tables** from known ones: 6 × 4 is double 3 × 4; 7 × n = 5 × n + 2 × n.
2. **Grid method:** 23 × 4 = (20 × 4) + (3 × 4) = 80 + 12 = 92; then the column method.
3. **Multiplying by 0 and 1.**
4. **Quick recall (5 min daily):** target 18 of 20 facts in under 3 seconds each.

Tables of 8 and 9 and division follow in Term 2.

Workbook pages

Page	Outcome	Activity
1–2	MA-3.1	Write numbers in the chart and in words; compare with <, >, =.
3	MA-3.1	Round to the nearest 10 and 100 on a number line.

Page	Outcome	Activity
4	MA-3.1	Odd or even sort; Roman numerals I–XX match.
5–7	MA-3.2	Estimate, then calculate: 8 additions, 8 subtractions (including across zeros).
8	MA-3.2	Three word problems (school library, cricket scores, shop).
9–10	MA-3.3	Tables of 6 and 7 grid; grid-method multiplications.
11	MA-3.3	Two-step word problem.

Question bank

MA-3.1

#	Question	Your answer
1	In 4,732 the 7 means ...	7 ones / 7 tens / 7 hundreds / 7 thousands
2	Round 4,732 to the nearest 10.	4,700 / 4,730 / 4,740 / 4,800
3	Round 4,732 to the nearest 100.	4,700 / 4,730 / 4,800 / 5,000
4	Which sign makes this true: 3,509 __ 3,590?	< / > / =
5	Which number is even?	47 / 63 / 58 / 91
6	IX in Roman numerals is ...	11 / 9 / 6 / 4
7	Write in figures: six thousand and forty.	_____
8	2,999 + 1 = 3,000. True or false.	True / False
9	Which is the biggest?	7,099 / 7,909 / 9,077 / 9,007
10	XV in Roman numerals is ...	5 / 10 / 15 / 20

MA-3.2

#	Question	Your answer
1	3,456 + 2,378 =	5,724 / 5,834 / 5,844 / 6,834
2	6,000 − 2,345 =	3,665 / 4,345 / 3,655 / 4,655
3	5,204 − 1,786 =	3,528 / 3,418 / 4,418 / 3,482
4	47 + 38 =	75 / 85 / 95 / 84
5	Estimate 3,890 + 2,120 to the nearest 100.	5,000 / 6,000 / 7,000 / 6,100
6	1,250 + 1,250 = 2,500. True or false.	True / False
7	A library has 2,450 books. It buys 375 more. How many now?	_____
8	Team A scored 1,206 runs and Team B 987. How many more did A score?	_____
9	92 − 45 =	57 / 47 / 53 / 43

#	Question	Your answer
10	To check 700 − 250 = 450, we can add 450 + 250. True or false.	True / False

MA-3.3

#	Question	Your answer
1	6 × 7 =	36 / 42 / 48 / 49
2	7 × 8 =	54 / 56 / 63 / 48
3	6 × 9 =	54 / 56 / 45 / 63
4	7 × 7 =	42 / 49 / 56 / 77
5	23 × 4 =	82 / 92 / 27 / 96
6	46 × 3 =	128 / 138 / 148 / 49
7	Any number × 0 = 0. True or false.	True / False
8	15 × 6 =	80 / 90 / 96 / 21
9	A box holds 6 eggs. How many eggs in 7 boxes?	_____
10	Pencils cost Rs 24 each. How much for 3 pencils and a Rs 10 eraser?	_____